RAJAT MURHE

Final Year Computer Science Undergraduate
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PROFILE

Final year Computer Science undergraduate (BTech, AI and Machine Learning, VIT Chennai) with three internships, who builds **voice and clinical AI**: a voice receptionist that books real appointments and refuses medical advice, and an explainable diagnostic model for Alzheimer's stages. I care about safety checks, multilingual conversations in Hindi, English and Marathi, and finding the failures that only show up in live testing.

EXPERIENCE

AI Product Growth and Strategy Intern, DCS AI Technologies Sep 2026 to Present

- Integrated **third party APIs** into AI powered product workflows, connecting external services with application features and validating end to end functionality.
- Worked on a platform's multimodal AI capabilities, testing and validating image, video, audio and product ad generation workflows, including API responses and **credit consumption**.
- Contributed to an AI connector platform's integration workflows by testing connectors, dashboard operations, execution flows and **role based permissions**.
- **Debugged** integration issues and documented test cases and findings to improve product reliability and release readiness.

AI/ML Intern, Linkific | Remote Jun 2026 to Aug 2026

- Built the **backend** for a Secure Enterprise AI Assistant that answers questions from company documents, using FastAPI, RAG on PostgreSQL with pgvector and a Groq LLM.
- Added **role based access control** to the API so people only get answers from documents they are allowed to see, and helped with troubleshooting and debugging for users.

AI Intern, Rungta Ventures AI Jan 2026 to May 2026

- Built a **semantic search** feature on MyCA using embeddings, and wrote Python scripts with NLTK and spaCy that validate uploaded documents, with a team of around ten people on a live product that uses Firebase for OTP login.

PROJECTS

ClinicVoice AI, Voice Receptionist for Clinics | GitHub Jun 2025 to Sep 2025

Python* | *FastAPI* | *Node.js* | *LangGraph* | *MCP* | *Groq* | *Amazon Bedrock* | *SQLite* | *Docker Compose* | *GitHub Actions

- Built a voice receptionist for clinics that listens with Whisper, answers patient questions in Hindi, English and Marathi, speaks back with XTTS v2, and books real appointments through **Google Calendar**.
- Added four safety layers so it refuses medical advice, and a **global safety gate** after live testing showed that the booking, refill and insurance agents had no safety check of their own.
- Added semantic caching, which in project testing cut average response time from about 8.1 seconds for a cold response to 48 milliseconds for a cached one, and ran an eval harness with a red team set in CI.

NeuroAI, Explainable Clinical Diagnostic Platform | GitHub | Live

Python* | *FastAPI* | *Scikit learn* | *XGBoost* | *PyTorch* | *BERT* | *SHAP

- Built a web app that predicts Cognitively Normal, MCI or Alzheimer's Disease from 8 patient measures. Random Forest and XGBoost scored highest at **85.23% accuracy**, with a fine tuned BERT at 81.82%, and SHAP explains each prediction.

Atlas, AI Powered Commercial Intelligence Workbench | GitHub

Python* | *SQL* | *PostgreSQL* | *LangGraph* | *FastAPI* | *Next.js* | *TypeScript* | *Redis* | *Docker* | *GitHub Actions

- Developed a **conversational analytics agent** in LangGraph that keeps context across questions, validates the SQL it writes, and answers with evidence from a PostgreSQL database of 24,576 synthetic pharmaceutical sales records.
- Built forecasting that reached 1.25% forecast MAPE versus 4.70% for a seasonal naive baseline, and a Next.js dashboard with interactive charts, with 19 backend tests in GitHub Actions.

FieldProof AI, Field Service Video QA Platform | GitHub Aug 2026 to Sep 2026

Python* | *FastAPI* | *Next.js* | *TypeScript* | *PostgreSQL* | *Gemini* | *Clerk

- Built a multi tenant platform where **Gemini** checks technician job videos against a checklist and a person reviews and approves each finding, with an audit trail of every job step.
- Designed a PostgreSQL **job queue** with retries and stale job recovery, so long video checks run on their own without the frontend open.

TECHNICAL SKILLS

Voice and Conversational AI: Whisper, XTTS v2, LangGraph, MCP, Groq, Ollama, Amazon Bedrock, safety guardrails, semantic caching, multilingual assistants (Hindi, English, Marathi), Google Calendar API

Healthcare and Machine Learning: Python, PyTorch, BERT fine tuning, Scikit-learn, XGBoost, SHAP, Hugging Face, NLTK, spaCy, model evaluation

Backend and Deployment: FastAPI, Node.js, PostgreSQL, SQLite, Docker Compose, GitHub Actions CI, Git, React, Next.js, TypeScript

CS Fundamentals: Data structures and algorithms (DSA), DBMS, object oriented programming (OOPs), Python, Java, C++, Go

EDUCATION

Vellore Institute of Technology, Chennai | BTech, Computer Science, Specialization in AI and Machine Learning Aug 2023 to Sep 2027

CERTIFICATIONS AND ACHIEVEMENTS

- Won **First Prize** at the AGRHINOVA Hackathon, VIT Chennai, for a crop diagnostics platform that combines AI image classification with a multilingual RAG assistant.
- AWS Certified Developer Associate. AWS Cloud Practitioner. Google Cloud Professional ML Engineer. Microsoft Azure Developer Associate, Azure Fundamentals, ML Operations Engineer Associate. IBM Advanced Generative AI. Avery Dennison Scholarship.

LEADERSHIP AND EXTRACURRICULAR

- Vice President, Swarajya Marathi Literary Association (2025 to 2026). State Level Cricketer, Maharashtra Under 16.